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Accessing Gyroscope Data on Nano 33 IoT

Accessing Gyroscope Data on Nano 33 IoT

Learn how to measure the direction of force to emulate an object's crash using the Nano 33 IoT board.

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This tutorial will focus on the 3-axis gyroscope sensor of the **LSM6DS3** module, on the Arduino Nano 33 IoT, in order to measure the direction of force on the board to emulate an object's crash. This will be achieved by utilizing the values of the gyroscope's axes and later print the return values through the Arduino IDE Serial Monitor.

Goals

The goals of this project are:

- Understand what an LSM6DS3 module is.

- Use the LSM6DS3 library.

- Read data from the gyroscope's sensor.

- Convert the raw data into direction of force.

- Print out live data through the Serial Monitor.

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Goals

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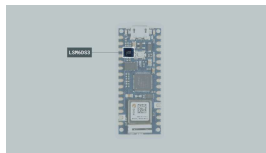
Hardware & Software Needed

This project uses no external sensors or components.

In this tutorial we will use the Arduino Create Cloud Editor to program the board.

The LSM6DS3 Inertial Module

IMU stands for: inertial measurement unit. It is an electronic device that measures and reports a body's specific force, angular rate and the orientation of the body, using a combination of accelerometers, gyroscopes, and oftentimes magnetometers. In this tutorial we will learn a bit more of the LSM6DS3 IMU module, which is included in the Arduino Nano 33 IoT Board.



The LSM6DS3 sensor.

The LSM6DS3 is a system-in-package featuring a 3D digital linear acceleration sensor and a 3D digital angular rate sensor.

The Library

The Arduino LSM6DS3 library allows us to use the Arduino Nano 33 IoT IMU module without having to go into complicated programming. The library takes care of the sensor initialization and sets its values as follows:

Accelerometer range is set at -4 | +4 g with +/-0.122 mg resolution.

Gyroscope range is set at -2000 | +2000 dps with +/-70 mdps resolution.

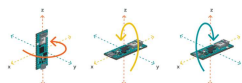
Output data rate is fixed at 104 Hz.

If you want to read more about the LSM6DS3 sensor module see [here](#).

Gyroscope

A gyroscope sensor is a device that can measure and maintain the orientation and angular velocity of an object.

Gyroscopes are more advanced than accelerometers, as they can measure the tilt and lateral orientation of an object, whereas an accelerometer can only measure its linear motion.



How gyroscopes work.

Gyroscope sensors are also called "Angular Rate Sensors" or

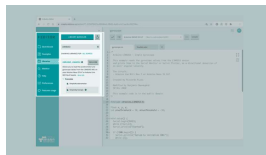
second, angular velocity is the change in the rotational angle of the object per unit of time.

In this example, we will use the gyroscope as an indicator for the direction of the force that is applied to the board. This will be achieved by swiftly moving the board for an instant in four directions: forward, backward, to the left and to the right. The results will be visible through the Serial Monitor.

Creating the Program

1. Setting up

Let's start by opening the [Arduino Cloud Editor](#), click on the **Libraries** tab and search for the **LSM6DS3** library. Then in **> Examples**, open the **SimpleGyroscope** sketch and once it opens, rename it as **Gyroscope**.

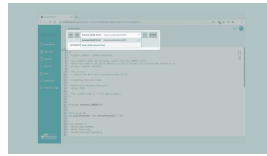


Finding the library in the Cloud Editor.

2. Connecting the board

Now, connect the Arduino Nano 33 IoT to the computer and make sure that the Cloud Editor recognizes it, if so, the board and port should appear as

they don't appear, follow the [instructions](#) to install the plugin that will allow the Editor to recognize your board.



Selecting the board.

3. Printing the board's direction of force

Now we will need to modify the code on the example, in order to print the board's direction of force as we move it in different angles.

Let's start by initializing the the x, y, z axes as `float` data types, and the

```
int plusThreshold = 30;
```

 and

```
int minusThreshold = -30;
```

value threshold variables before the `setup()`.

In the `setup()` we should **remove** the last line of code, as we won't print the raw values of the axes:

```
1 Serial.println("X\tY\tZ\t");
```

Next, in the `loop()` we can **remove** the following, since we initialized in the beginning,

```
1 float x, y, z;
```

as well as the following lines
that won't be required:

```
1 Serial.print(x);  
2 Serial.print('\t');  
3 Serial.print(y);  
4 Serial.print('\t');  
5 Serial.println(z);
```

Instead, we will add four `if` statements, following the already existing one that checks if the sensor is available and then reads the 3 axes.

Here is the code we will add:

```
1 if(y > plusThreshold)  
2 {  
3   Serial.println("Colli  
4   delay(500);  
5 }  
6 if(y < minusThreshold  
7 {  
8   Serial.println("Colli  
9   delay(500);  
10 }  
11 if(x < minusThreshold  
12 {  
13   Serial.println("Colli  
14   delay(500);  
15 }  
16 if(x > plusThreshold)  
17 {  
18   Serial.println("Colli  
19   delay(500);  
20 }
```

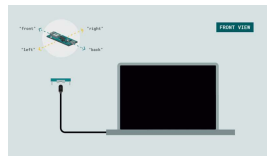
These `if` statements will check the x and y values of `plusThreshold` and `minusThreshold` that we defined in the beginning. According to these values the direction of the force that we move our board towards, will be printed.

4. Complete code

If you choose to skip the code building section, the complete code can be found below:

```
1  /*
2   Arduino LSM6DS3 -
3
4   This example reads
5   and prints them to
6   an axis' angular v
7
8   The circuit:
9   - Arduino Uno WiF:
10
11   Created by Riccaro
12
13   Modified by Benjar
14   30 Nov 2020
15
16   This example code
17  */
18
19
20  #include <Arduino_L3
21
22
23  float x, y, z;
24  int plusThreshold =
25
26
27  void setup() {
28    Serial.begin(9600);
```

In order to get a correct reading of the board data, before uploading the sketch to the board hold the board in your hand, from the side of the USB port. The board should be facing up and "pointing" away from you. The image below illustrates the board's position and how it works:

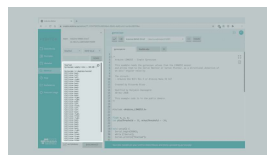


Positioning of the board.

Next, you can verify and upload the sketch to the board and open the Monitor from the menu on the left.

Now with the board parallel to the ground you can swiftly move it towards one direction: forward, backwards, right or left. According to the movement of your choice, the results will print every second to your monitor!

Here is a screenshot of the sketch returning these values:



Serial Monitor output.

Troubleshoot

Sometimes errors occur, if the

some common issues we can
troubleshoot:

Missing a bracket or a
semicolon.

Arduino board connected
to the wrong port.

Accidental interruption of
cable connection.

Conclusion

In this simple tutorial we
learned what an IMU sensor
module is, how to use the
LSM6DS3 library, and how to
use an Arduino Nano 33 IoT
microcontroller, to measure and
print out values from a
gyroscope sensor. Furthermore,
we created an application that
detects the direction of force
that we can apply to the board.

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